# Hub-Aware Incremental Fraud Ring Detection on Streaming Transaction Graphs

Roshan Parveen
*School of Computer Science and Engineering*
*Vellore Institute of Technology*
Chennai, India
roshan.parveen2023@vitstudent.ac.in

Sanjay Kumar S
*School of Computer Science and Engineering*
*Vellore Institute of Technology*
Chennai, India
sanjaykumar.s2023@vitstudent.ac.in

***Abstract*—Fraud rings share devices, cards and network addresses and move money among themselves, which graph queries can expose. Two obstacles appear on live transaction streams. Legitimate shared infrastructure such as office gateways, carrier address translation and kiosks links many unrelated accounts, and the usual remedy, a hard degree cap, hides any ring that shares a busy attribute. Whole-graph queries re-run on a timer also cost more as the graph grows. We replace the cap with a continuous hub-aware weight with time decay, detect incrementally on the neighbourhood touched by each micro-batch, and pass candidate rings through a calibrated scorer that explains every alert. On a deterministic simulator with ring-level ground truth and five evasion strategies, over 5 scenarios and 10 test seeds, the full pipeline reaches a ring F1 score of 0.73 against 0.21 for the original degree-capped Cypher rules. Most of that gap is the scorer: with the same calibrated scorer on top, the capped rules reach 0.56 and Louvain communities 0.41, and the remaining differences are significant in a paired, Holm-corrected Wilcoxon test. Keeping links through busy attributes instead of capping them extends the reachable recall on shared-attribute rings from 0.53 to 0.81. The particular degree discount matters little: with its own tuned threshold, a constant weight is as accurate (F1 0.76 against 0.73), while a soft cap at degree 10 loses most rings hidden behind hubs. Incremental detection flags the same rings as full recomputation and, with 100,000 users, takes 36.1 ms per batch against 22,501 ms. Rings that rotate identities or act slowly evade every method we tested. On Neo4j and Redis Streams the detector flags the same rings. A learned classifier over rule-free groups reaches F1 0.555 and collapses under decoy edges. Measuring explanation quality exposed counterfactuals that never changed the decision, which we fixed. All positive results are synthetic, and we state what the public anti-money laundering datasets we could use can and cannot test.**

***Index Terms*—fraud rings, property graphs, streaming graph analytics, incremental detection, hub-aware weighting, explainability**

## I. Introduction

Fraud rings are groups of accounts that act together: they share devices, payment cards or network addresses, and they move money among themselves in cycles or fan-in and fan-out patterns. Graphs make these relations explicit, which is why graph-based methods dominate fraud and anti-money laundering (AML) research [1], [2]. Two problems appear as soon as such methods run on a live stream of events.

**The hub dilemma.** Legitimate shared infrastructure, such as an office gateway, a carrier-grade network address translation (NAT) address or a store kiosk, links many unrelated accounts. Linking accounts by shared properties produces sparse links through strongly identifying entities and over-linkage through widely shared ones such as IP addresses [3]. The usual remedy in rule-based systems is a hard degree cap: attributes shared by more than $c$ accounts are ignored. The cap removes the noise but also opens an evasion path, because a ring that shares a busy attribute becomes invisible to it, much as fraudsters elsewhere hide by connecting to benign entities [4].

**Latency on a stream.** Accounts, links and transfers arrive continuously. Re-running whole-graph queries on a timer delays detection, and the cost of each pass grows with the graph rather than with the change, a problem that incremental community detection addresses for clustering [5], [6].

We make four contributions.

- **C1, hub-aware weighting.** We replace the degree cap with a continuous weight: each shared attribute contributes $1/\ln(1+d)$, where $d$ is its degree, with exponential time decay, and transfers inside the pair add weight. The discount itself is FRAUDAR's column weighting [7]; our contribution is its use for shared-attribute ring detection on a stream. Busy hubs count for little on their own, but a group that shares a hub and also trades among itself still accumulates weight (Section III). What we can show is that keeping hub links is what matters: a constant weight with its own threshold performs as well, and caps lose rings hidden behind hubs (Section V).
- **C2, incremental detection.** A micro-batch engine re-evaluates only the $k$-hop region touched by each batch, with fan-out limits that keep that region bounded when it reaches a hub.
- **C3, calibrated, explained alerts.** Candidate rings are scored by a calibrated classifier, and each alert carries feature contributions, an evidence subgraph and counterfactuals. The scorer supplies most of the precision, for our candidates and for the baselines' (Section V).
- **C4, evaluation under evasion.** A deterministic simulator with ring-level ground truth and five evasion strategies, including an attacker that adapts to the detector's flags, lets us measure each component on identical event streams (a paired design) with confidence intervals.

## II. Related Work

**Graph fraud detection.** Most recent graph fraud detection work trains graph neural networks (GNNs) to label nodes. It addresses heterophily, where fraudulent nodes connect mostly to benign ones [8], [9]; camouflage, by reweighting neighbours adversarially [4]; scarce or unreliable labels, with one-class and semi-supervised training [10]–[12]; and interpretability [13]. These methods output per-node scores and need labelled training graphs. GB-FRD [2] targets fraud rings in peer-to-peer payment graphs, as we do, and Yin et al. [3] replace hard shared-property links with behavioural similarity because of over-linkage through shared entities. We keep shared-attribute links, which are cheap and explainable, and change how much each one counts.

**Dense subgraphs and peeling.** Finding unusually dense groups is a classical route to fraud and collusion detection: greedy peeling approximates the densest subgraph [14], spectral and greedy densest-subgraph algorithms scale to billion-edge graphs [15], and sequential peeling of a weighted bipartite graph exposes coordinated adversarial annotators [16]. FRAUDAR [7] makes peeling resistant to camouflage by weighting each edge by a decreasing function of the degree of the object it points to, recommending $1/\log(x+c)$ and arguing that $1/x$ shrinks too fast; Spade [17] maintains such peeling-based dense subgraphs incrementally under edge insertions on transaction graphs. Core decomposition, the peeling primitive we use to extract dense cores, has efficient parallel and GPU implementations [18], [19] and is related to densest-subgraph approximation [20]. Our degree discount is therefore not new: it is FRAUDAR's column weighting, and close to Adamic–Adar. What we study is applying it to shared-attribute ring detection on a stream, as a replacement for the degree cap of rule-based systems, with incremental detection, ring tracking and calibrated alerts around it.

**Community detection on dynamic graphs.** Louvain modularity optimisation [21] remains a standard way to find groups and is still being refined [22]; we use it as a baseline. Incremental community detection avoids repartitioning a changing graph by updating only the affected communities [5], [6]. Community structure, however, is not ring structure: a benign hub joins thousands of users into one community.

**Similarity weighting.** Local similarity indices such as Jaccard and Adamic–Adar [23], which discounts a shared neighbour by the logarithm of its degree, are standard in link prediction [24]. Our hub weight applies a similar discount to shared attributes, and we match flagged rings to true ones by Jaccard overlap.

**AML on public data.** Work on the Elliptic Bitcoin graph compares tabular ensembles with GNNs under temporal splits [25]–[28]; AMLworld provides synthetic laundering attempts with typology labels [29]. These studies classify transactions, so their scores are not comparable with our ring-level metrics (Section V-F).

**Calibration and explanations.** Post-hoc calibration by isotonic regression [30] and its comparison with Platt scaling [31] make classifier scores usable as probabilities; Azbergen et al. [32] compare such calibrators for AML risk scores, and learned calibration has been studied beyond classification [33]. Counterfactual explanations state what would have to change for a different decision; their robustness [34], consistency across model updates [35] and per-feature importance [36] are active topics. We attach simple, model-specific counterfactuals to every alert.

## III. Method

Fig. 1 shows the system. A seeded simulator emits observable events (account joins and leaves, attribute links, transfers); ground truth goes only to a ledger the detector never sees.

### A. Graph model

Users link to devices, cards and IP addresses (`USED_DEVICE`, `HAS_CARD`, `LOGGED_IN_FROM`) and send transfers to each other. The same contract is implemented over Neo4j (Cypher) and over an in-memory store with identical semantics; all experiments below use the in-memory store. In live runs, events reach the engine through an in-process queue or Redis Streams, whose consumer settings (batch size, blocking timeout) trade throughput against latency [37].

### B. Hub-aware weighting (C1)

For two users $u, v$ the edge weight is

$$W(u,v) = \sum_{a \in A(u) \cap A(v)} h(\deg a)\, \delta(t_a) + \min\Big(\beta \sum_{x \in T(u,v)} \delta(t_x),\, \beta_{\max}\Big), \tag{1}$$

where $h(d) = 1/\ln(1+d)$, $\delta(t) = 2^{-t/\tau}$ is exponential decay with half-life $\tau = 300\,\mathrm{s}$, $t_a$ is the age of the older of the two links to attribute $a$, and $T(u,v)$ are the transfers between $u$ and $v$ in the window, with $\beta = 0.15$ and $\beta_{\max} = 0.6$. A device shared by three accounts weighs $0.72$; an office address shared by fifty weighs $0.25$. Candidate rings are the connected components of the graph of pairs with $W \geq \theta_e$ (and a non-zero attribute term), peeled to cores of edge density at least $0.75$ and size 3–25. The edge threshold $\theta_e$ is selected on validation seeds (Section IV-B).

### C. Incremental detection (C2)

Events are grouped into micro-batches of 50. For each batch the engine takes the users it touched, expands them by $k = 2$ hops through shared attributes and transfers, and builds a view of that region over a 900 s window, together with the *global* degree of every attribute in it, so $h(\deg a)$ is the same as in a full pass. The first hop may cross attributes with up to 250 users (so the co-members of a hub are re-examined); later hops cross only attributes with at most 20 users, which stops the region from spreading through hubs. The first-hop limit is itself a degree cap on the region: a pair whose only link is an attribute shared by more than 250 users never enters the same region. For the hub-aware weight this costs nothing, since one such attribute contributes at most $1/\ln 252 \approx 0.18$,

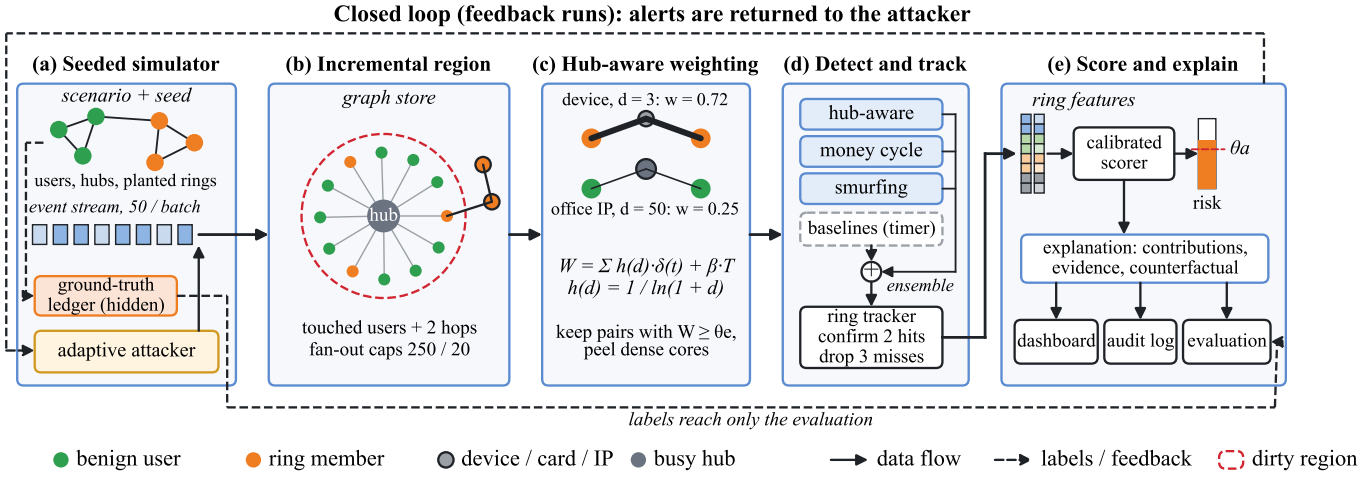



Fig. 1. Overview of the system. (a) A seeded simulator emits the observable event stream in micro-batches; labels go only to a ledger. (b) Each batch is written to the graph store and expanded to the bounded region it touched. (c) Shared attributes are weighted by their degree, so a busy hub counts for little and a rarely shared device for much. (d) Incremental detectors run on the region and a tracker confirms rings across batches. (e) Confirmed rings are scored, explained and alerted. Solid arrows: data flow; dashed arrows: labels (evaluation only) and, in closed-loop runs, alerts returned to the attacker.

below the edge threshold, so a full pass would not link the pair either; a weight that lets one busy attribute reach the threshold, such as a constant, does lose such pairs (Section V). Incremental detectors (hub-aware shared attribute, temporal cycle, smurfing) run on the view; whole-graph detectors (the static Cypher rules and community detection) run on a timer.

### D. Ensemble, tracking and scoring (C3)

Candidates from different detectors are merged when their Jaccard overlap is at least 0.5, or when one contains 80% of the other and they differ in size by at most a factor of two. A tracker matches candidates to existing rings across batches; a ring is *confirmed* after two consecutive hits and dissolved after three misses. Confirmed rings are scored by a logistic-regression model over 19 ring features (size, density, which attribute kinds are shared, the hub-adjusted weight, cycle presence and length, burstiness, account ages, internal transfers per member, amount statistics, growth, hub exposure and the number of agreeing detectors), calibrated with isotonic regression [30]. A ring becomes an alert when its calibrated risk reaches the alert threshold $\theta_a$. Each alert carries the per-feature contributions to the score, the subgraph of shared attributes and transfers that formed it, and counterfactuals (for the leading features, the value at which the risk would fall below the threshold, and for hub-aware rings the degree the shared attribute would need before the rule stops firing). The model is trained on seeds disjoint from all evaluation seeds (Fig. 2): the detector runs on six scenarios with seeds 101–110, each confirmed or changed ring becomes a row of its 19 features, labelled from the hidden ledger, and within each run the first 70% of rows fit the regression and the last 30% the calibrator.

TABLE I
SCENARIOS (FROM `CONFIGS/SCENARIOS/*.YAML`).

| Scenario | Users | Dur. (s) | Ev./s | Rings and evasion |
|---|---|---|---|---|
| baseline | 30 | 60 | 10 | 4 planted, none |
| high_noise | 150 | 60 | 50 | 4 planted + random, decoys |
| hub_heavy | 100 | 60 | 25 | 4 planted, hub hiding |
| adversarial | 120 | 90 | 30 | 5 planted + random, all five |
| drift | 200 | 90 | 40 | 2 planted + random, hub hiding |
| demo_story | 120 | 90 | 20 | 3 planted, hub hiding |
| scale_sweep | $10^3$+ | 120 | 100 | latency benchmark (E4) |

## IV. Experimental Setup

### A. Simulator and scenarios

The simulator is a deterministic discrete-event generator: a scenario file and a seed fix the whole stream, which we verify by a digest of the event sequence recorded with every result. Benign users join and leave, link to devices, cards and IPs drawn from heavy-tailed popularity, and transact; benign hubs (offices, carrier NAT, kiosks, family devices) are generated with configured sizes. Ring types are shared device, card or IP, mixed attribute, money cycle, layered chain and smurfing fan-in/fan-out. Evasion strategies are hub hiding (the ring shares a busy benign hub), low-and-slow activity, identity rotation, decoy edges to benign users, and an adaptive attacker that shifts its strategy mix away from strategies that were flagged. Fig. 3 shows the generator and its two outputs, and Table I lists the scenarios.

### B. Protocol

Seeds are split three ways: scorer training (101–110, test 201–205), threshold selection (validation, 301–303) and evaluation (test, 1001 and up). On the validation seeds we evaluated a grid of alert thresholds $\theta_a \in \{0.2, 0.3, 0.5, 0.7, 0.8, 0.9\}$

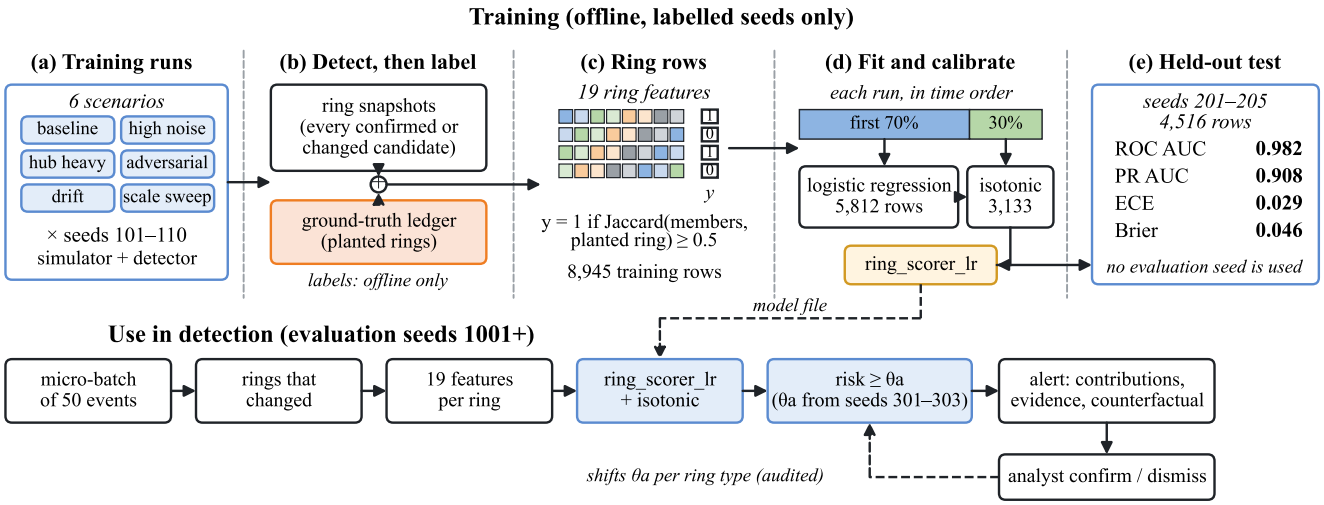



Fig. 2. Training and use of the ring scorer. (a)–(b) The simulator and detector run on training seeds, and ring snapshots are labelled from the ground-truth ledger, which the detector never reads. (c) Each snapshot is a row of 19 ring features. (d) Within each run, rows in time order are split 70/30 between logistic regression and isotonic calibration. (e) Held-out test seeds (values from `models/ring_scorer_lr.json`). Below: in detection, each changed ring is scored every batch, and an alert carries its explanation; analyst feedback adjusts the threshold.

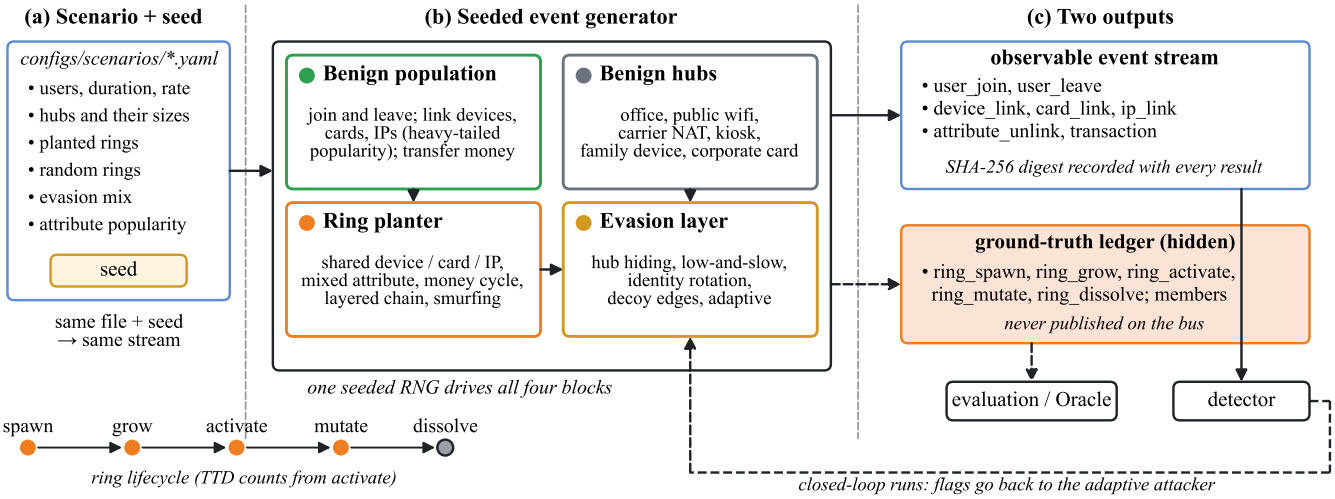



Fig. 3. Simulator design. (a) A scenario file and a seed fix the whole run. (b) One seeded generator combines a benign population, benign hubs, planted rings and an evasion layer. (c) Observable events go to the detector, with a digest of the sequence recorded with every result; ring lifecycle events go only to the ground-truth ledger used for evaluation and the Oracle view. In closed-loop runs, flags are returned to the adaptive attacker.

and edge thresholds $\theta_e \in \{0.2, 0.3, 0.4, 0.5, 0.6\}$ on the five accuracy scenarios; the grid was widened whenever the best value sat on its edge. The selected values are $\theta_a = 0.8$ and $\theta_e = 0.3$ (mean validation F1 0.72). We report the proposed method both with these values and with the a-priori values $\theta_a = \theta_e = 0.5$.

Every method in an experiment processes the *identical* event stream for a given scenario and seed (the attacker does not receive feedback in these runs), so differences between methods are paired. We report the mean and 95% $t$-interval over runs, and test differences with the Wilcoxon signed-rank test on paired runs, with Cohen's $d_z$ as effect size.

### C. Metrics

A flagged ring *matches* a true ring when their member sets overlap with Jaccard at least 0.5. Ring precision is the fraction of flagged rings that match a true ring; ring recall is the fraction of true rings matched at some point while active. A run in which nothing is flagged has undefined precision; we count its F1 as 0. Time to detect (TTD) is the time from a ring's activation to its first matching flag. Detection latency is the wall-clock time per micro-batch for the dirty-region expansion, the view and the incremental detectors.

### D. Baselines

- **Static Cypher, cap 100 / cap 10.** The original rules: shared device (Q1), shared card (Q2), 3-cycles of transfers (Q3), and shared IP with a degree cap (Q4a/Q4b), run on the whole graph every 5 s of simulated time.
- **Louvain.** Louvain communities [21] of the user graph projected through shared attributes and transfers, kept when dense enough.
- **Connected components.** Weakly connected components of the same projection, no density filter.

All baselines use the same two-hit confirmation as the proposed method.

### E. Environment

All runs used one laptop (Intel Core i3-1115G4, 2 cores / 4 threads, 7.7 GB RAM, Windows 11), Python 3.11.9, NumPy 2.2.6, scikit-learn 1.7.2, NetworkX 3.5, with experiments run one at a time.

## V. Results

### A. RQ1: accuracy against baselines

Table II pools 5 scenarios × 10 test seeds for every method, all on identical streams. Without a scorer, the static rules reach ring F1 0.21 with a degree cap of 10 and 0.19 with a cap of 100, Louvain 0.29 and connected components 0.00. These rows differ from the proposed method in two ways at once: how candidates are generated and whether they are scored. With the same calibrated scorer and an alert threshold selected per method on the validation seeds, the static rules reach 0.56 (cap 10) and 0.52 (cap 100) and Louvain 0.41, against 0.73 ($\pm$0.05) for the proposed method; every difference to the proposed method remains significant after Holm correction ($p < 0.001$). Within the scored group the differences isolate the candidate generator, the hub-aware rule and incremental detection; between a generator's unscored and scored rows they isolate the scorer. For the rules with cap 10, the scorer raises F1 from 0.21 to 0.56, through precision (0.14 to 0.64) at some cost in recall (0.61 to 0.53); the candidate generator raises it further to 0.73, mostly through recall (0.53 to 0.74). The scorer was trained on candidates of the proposed pipeline, so on rule and Louvain candidates it works under some distribution shift. The unscored static rules reach recall 0.61 but precision only 0.14: the shared-attribute rules and the 3-cycle rule fire on many benign groups, and two-hit confirmation does not remove them. Connected components never produce a group within the ring size limits, because benign hubs join almost all users into one component; Louvain splits that component, but its communities rarely coincide with a ring. The thresholds selected on validation seeds improve on the a-priori ones (F1 0.73 vs. 0.68, $p = 0.03$).

Per scenario (Table III), the gap to the static rules is largest on `drift` and `high_noise`, where many benign groups satisfy the rules and their precision collapses, and smallest on the clean `baseline` scenario and on `adversarial`, which is the hardest scenario for every method.

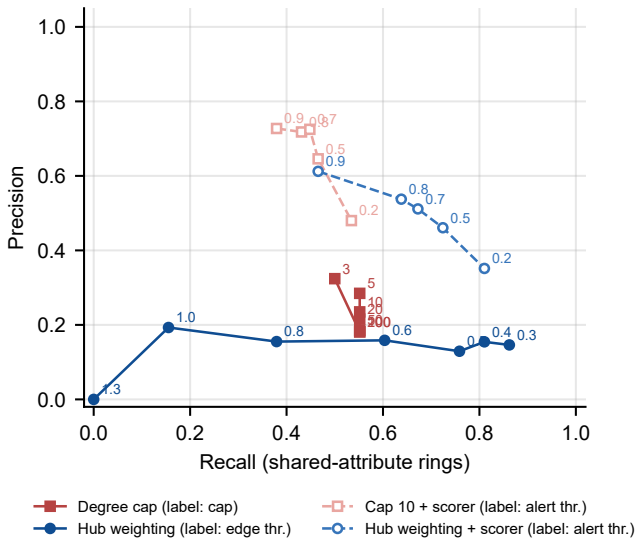



Fig. 4. Precision and recall on shared-attribute rings (micro-averaged over 4 scenarios × 5 test seeds). Solid: detector alone, swept over the degree cap or the edge threshold $\theta_e$. Dashed: with the calibrated scorer, swept over the alert threshold $\theta_a$ (cap 10; $\theta_e = 0.3$). Best-F1 difference with the scorer (hub minus cap): 0.03, 95% bootstrap interval -0.03 to 0.10 (`results/e3b_frontier_ci`).

**Degree cap vs. hub weighting.** Fig. 4 isolates the mechanism this paper replaces. Only shared-attribute and mixed rings count as ground truth here (flags that match only other ring types are ignored), the static rules run without the cycle rule, and the hub-aware detector runs alone. Without the scorer, no degree cap reaches a recall above 0.55: rings that share a busy attribute are invisible to every cap. Hub weighting reaches recall 0.86, but on its own at precision 0.15. With the calibrated scorer on top, hub weighting reaches recall 0.81 (precision 0.35) and its best F1 is 0.58 (alert threshold 0.8); the cap with the scorer stays below recall 0.53 and its best F1 is 0.55. The difference in best F1, 0.03, has a 95% bootstrap interval of -0.03 to 0.10 over the 20 runs (the best threshold re-selected in every resample), so at the best operating point the two mechanisms are not distinguishable on F1. At low recall the cap with the scorer is the more precise of the two (0.72 at recall 0.45). Hub weighting therefore does not dominate the cap everywhere; it extends the frontier to recall levels the cap cannot reach, and the scorer is what makes those levels usable.

### B. RQ2: latency and scale

Table IV and Fig. 5 grow the user population of the `scale_sweep` scenario at a fixed event rate (100 events/s) and compare incremental detection with recomputing the same detectors on the whole graph for every batch. Full recomputation grows at least linearly with the graph (faster between 50,000 and 100,000 users, where the 7.7 GB test machine was short of memory; we did not separate the two effects). The incremental region stays between a few hundred and two thousand users, because the fan-out limits stop the expansion at hubs; it even shrinks as the graph grows, since this scenario

TABLE II

RING-LEVEL DETECTION ON 5 SCENARIOS × 10 TEST SEEDS, POOLED (MEAN ± 95% CI OVER RUNS; A RUN WITH NO FLAGS HAS UNDEFINED PRECISION AND COUNTS AS F1 = 0). TTD: MEDIAN TIME FROM RING ACTIVATION TO FIRST FLAG. LATENCY: MEDIAN PER-BATCH DETECTION TIME ON THE IN-MEMORY STORE; †FULL-GRAPH METHODS RUN ON A 5 S TIMER, NOT PER BATCH, SO THEIR ENTRY IS THE MEDIAN TIME OF ONE FULL-GRAPH PASS. $p$: PAIRED WILCOXON SIGNED-RANK TEST OF F1 AGAINST THE PROPOSED METHOD ON IDENTICAL STREAMS. SCORED BASELINES USE THE SHIPPED SCORER (`MODELS/RING_SCORER_LR.JSON`), WITH THE ALERT THRESHOLD SELECTED PER METHOD ON THE VALIDATION SEEDS; THE PROPOSED METHOD INCLUDES THE SAME SCORER, SO ONLY ROWS WITHIN A GROUP ISOLATE THE CANDIDATE GENERATOR. $p$ VALUES ARE HOLM-CORRECTED.

| Method | Precision | Recall | F1 | TTD (s) | Latency (ms) | $p$ |
|---|---|---|---|---|---|---|
| *Without a scorer* | | | | | | |
| Static Cypher (cap 100) | 0.12 ± 0.03 | 0.61 ± 0.08 | 0.19 ± 0.05 | 9.5 | 6.0† | <0.001 |
| Static Cypher (cap 10) | 0.14 ± 0.04 | 0.61 ± 0.08 | 0.21 ± 0.05 | 9.5 | 5.9† | <0.001 |
| Louvain communities | 0.39 ± 0.09 | 0.33 ± 0.08 | 0.29 ± 0.06 | 18.8 | 38.4† | <0.001 |
| Connected components | – | 0.00 ± 0.00 | 0.00 ± 0.00 | – | 11.1† | <0.001 |
| *With the calibrated scorer* | | | | | | |
| Static Cypher (cap 100) + scorer | 0.63 ± 0.08 | 0.48 ± 0.09 | 0.52 ± 0.08 | 8.6 | 4.5† | <0.001 |
| Static Cypher (cap 10) + scorer | 0.64 ± 0.07 | 0.53 ± 0.09 | 0.56 ± 0.07 | 9.3 | 4.6† | <0.001 |
| Louvain communities + scorer | 0.81 ± 0.12 | 0.32 ± 0.09 | 0.41 ± 0.09 | 19.0 | 27.9† | <0.001 |
| Hub-aware incremental (proposed) | 0.76 ± 0.06 | 0.74 ± 0.06 | **0.73 ± 0.05** | 9.3 | 26.6 | – |
| Proposed, a-priori thresholds | 0.73 ± 0.06 | 0.69 ± 0.06 | 0.68 ± 0.05 | 9.0 | 22.7 | 0.034 |

TABLE III

RING F1 PER SCENARIO (MEAN ± 95% CI OVER 10 TEST SEEDS).

| Method | baseline | high_noise | hub_heavy | adversarial | drift |
|---|---|---|---|---|---|
| Static Cypher (cap 100) | 0.48 ± 0.07 | 0.06 ± 0.02 | 0.24 ± 0.04 | 0.14 ± 0.04 | 0.05 ± 0.02 |
| Static Cypher (cap 10) | 0.54 ± 0.08 | 0.07 ± 0.02 | 0.27 ± 0.05 | 0.14 ± 0.04 | 0.05 ± 0.02 |
| Louvain communities | 0.44 ± 0.09 | 0.26 ± 0.19 | 0.24 ± 0.09 | 0.14 ± 0.09 | 0.38 ± 0.16 |
| Connected components | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.00 ± 0.00 |
| Hub-aware incremental (proposed) | 0.99 ± 0.03 | 0.62 ± 0.08 | 0.79 ± 0.08 | 0.59 ± 0.14 | 0.67 ± 0.04 |
| Proposed, a-priori thresholds | 0.93 ± 0.07 | 0.66 ± 0.05 | 0.66 ± 0.12 | 0.53 ± 0.11 | 0.62 ± 0.09 |

TABLE IV

PER-BATCH DETECTION LATENCY (MEDIAN, MS; MEAN ± 95% CI OVER 3 SEEDS) AND USERS EXAMINED PER BATCH, INCREMENTAL VS FULL RECOMPUTATION, IN-MEMORY STORE.

| | Incremental | | Full | | |
|---|---|---|---|---|---|
| Users | Region | ms | Region | ms | Speed-up |
| 1,000 | 976 | 211.0 ± 142.9 | 1015 | 159.7 ± 64.4 | 0.8× |
| 5,000 | 1878 | 177.8 ± 46.7 | 5016 | 589.8 ± 67.3 | 3.3× |
| 20,000 | 1324 | 54.6 ± 37.1 | 20010 | 2683.7 ± 1062.0 | 49.2× |
| 50,000 | 1110 | 38.5 ± 38.2 | 50011 | 6181.6 ± 2576.4 | 160.4× |
| 100,000 | 827 | 36.1 ± 44.2 | 100013 | 22500.5 ± 10278.5 | 623.0× |

scales the number of devices, cards and IPs with the users (so an ordinary attribute is shared by fewer users) while hub sizes stay fixed. Its latency therefore stays flat: at 100,000 users a batch takes 36.1 ms incrementally against 22,501 ms in full (623×). At 1,000 users the region already covers almost the whole graph, and the incremental mode is slower than full recomputation because of the cost of expanding the region. On the accuracy scenarios, incremental and full recomputation flag exactly the same rings in every run (ablation *no incremental* in Table V), so restricting detection to the dirty region loses nothing there.

On the small accuracy scenarios one static full-graph pass is cheap (5.9 ms) and cheaper than one batch of the proposed detectors (26.6 ms), which build the hub-aware projection and peel it. The proposed detectors run per batch, the static rules only on a timer; their time to detect is similar here (9.3 s vs. 9.5 s) because the timer runs every 5 simulated seconds.

### C. RQ3: robustness to evasion

**Hub size.** In Fig. 6 the office hub grows from 5 to 200 accounts. Recall of the hub-hiding rings does not fall with hub size for any method (proposed 71/90, static cap 10 65/90 over all sizes). The static rules do not find these rings through the shared hub: with the 3-cycle rule switched off, cap 10 finds 30/90, because the rings also move money among themselves in short cycles. What separates the methods is precision, and so F1 (0.80 vs. 0.31).

**Hubs above the fan-out limit.** Fig. 6 stops at 200 accounts, below the 250-user first-hop limit. We therefore resized the office hub to 500, 2,000 and 10,000 accounts (5 seeds each) and compared the shipped engine with full recomputation and with a "bound" rule that does cross an over-limit attribute,

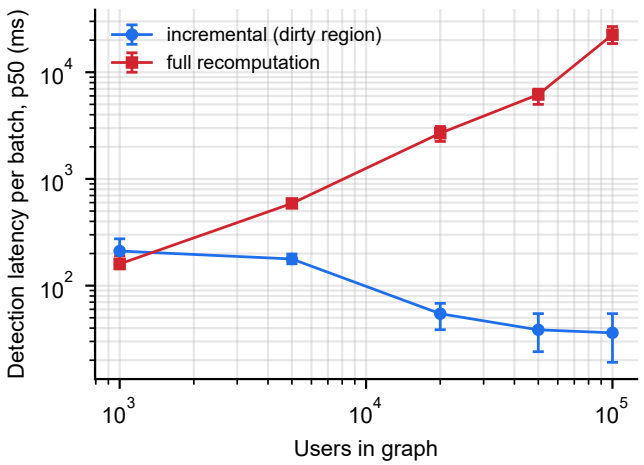



Fig. 5. Median per-batch detection latency as the graph grows (mean over 3 seeds; bars: per-seed minimum and maximum), incremental (dirty region) vs. full recomputation of the same detectors, in-memory store.

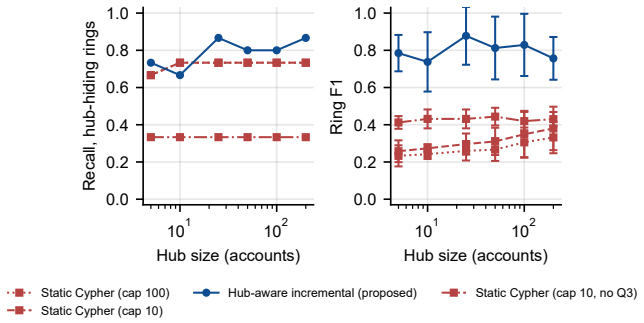



Fig. 6. A shared-IP ring hiding behind an office hub of growing size (`hub_heavy`, 5 test seeds per size). Left: recall of the hub-hiding rings; right: ring F1 (mean, 95% CI).

towards those co-members whose attribute weight with the touched user could reach the edge threshold. At every size the bound rule flagged the same rings as the 250-user limit, with identical precision, recall and region size (15/15 runs): it never added a user. Full recomputation found no more hub-hiding rings either (11/15 vs. 11/15 at 500, 9/15 vs. 10/15 at 2,000), while the incremental region stayed at a median of 14 users against up to 2,076 for full recomputation. The test does exercise the limit: on the same streams a constant weight (edge threshold 1.0, as selected in RQ5) loses hub-hiding rings to it (9/15 incremental vs. 12/15 full at 500; F1 0.74 vs. 0.85), at a cost of about 32 minutes per full run at 2,000 users. So, for hubs up to 10,000 accounts, the limit did not hide hub-sharing rings for the hub-aware weight; it does for a constant weight. The static rules stay well behind (F1 0.46, 0.60, 0.63).

**Evasion strategies.** Fig. 7 breaks recall down by the strategy each planted or random ring uses. The proposed method's advantage in recall comes mostly from rings without evasion (91/110 vs. 67/110 for cap 10) and rings with decoy edges (22/27 vs. 13/27). On hub-hiding rings the two are close (35/52 vs. 33/52), and identity rotation (4/14) and low-and-slow activity (3/13) defeat every method we tested, presumably because rotation removes the shared attributes the detectors rely on and slow rings accumulate little evidence inside one window.

**Adaptive attacker.** With feedback switched on, the attacker is told which of its rings were alerted and shifts its strategy mix. Over 10 test seeds it switched strategy 22 times and launched 22 adaptive rings, of which 18 were detected; without feedback, 9 of 11. Adaptation did not lower the detection rate in our simulator, although its strategies are limited to the four we implemented.

### D. RQ4: sensitivity

Raising the event rate from 10 to 100 events/s lowers the proposed method's F1 from 0.61 to 0.43 and raises its per-batch latency from 10.1 to 74.5 ms, since each batch touches more of the graph (Fig. 8). Larger planted rings are harder (F1 0.98 at size 3, 0.76 at size 8); a likely cause, which we did not isolate, is the 0.75 edge density the peeling requires. The view window between 60 and 900 s changes F1 little (0.64 vs. 0.67).

### E. RQ5: ablations

Table V removes one component at a time (20 paired runs). The calibrated scorer contributes most: without it F1 falls from 0.72 to 0.39 because precision drops to 0.28. Without the hub-aware detector, F1 falls to 0.36 and to 0.00 on `hub_heavy`. Removing the cycle or the smurfing detector costs 0.64 and 0.65. Replacing the hub weight by a constant at the same edge threshold collapses `hub_heavy` (0.81 to 0.42) but slightly helps `high_noise` (0.54 to 0.68), so over all four scenarios the difference is not significant ($p = 0.161$). That ablation keeps the edge threshold tuned for $1/\ln(1+d)$, which is unfair to the constant; the next paragraph gives each weight its own threshold.

**Weight functions.** We compared five weights $h(d)$, each with its edge threshold selected on the validation seeds and the alert threshold fixed at 0.8: the shipped $1/\ln(1+d)$ ($\theta_e = 0.3$), $1/d$ (0.1), $1/\sqrt{d}$ (0.2), a soft cap $1/(1+e^{(d-10)/2})$ (1) and a constant (1); for $1/d$, the soft cap and the constant the selected value lies on the edge of the grid. On `hub_heavy` with 30 paired seeds, the shipped weight reaches F1 0.79 and recall 0.87 of the rings hidden behind hubs; the constant reaches 0.81 and 0.83 ($p = 0.249$), $1/\sqrt{d}$ 0.79, while $1/d$ (0.56) and the soft cap (0.60, hidden-ring recall 0.27 at precision 0.99) are worse ($p < 0.001$). Over the five scenarios the picture is the same: 0.73 for the shipped weight, 0.76 for the constant and 0.74 for $1/\sqrt{d}$, none significantly different even before correction, and 0.62 for $1/d$ and 0.68 for the soft cap. Discounting busy attributes too strongly, or switching them off, loses rings that share a hub; within the range that keeps hub links, the shape of the discount makes no measurable difference once the edge threshold is tuned for it. Hysteresis and time decay make no significant difference here; without decay F1 is even slightly higher (0.75). The accuracy scenarios

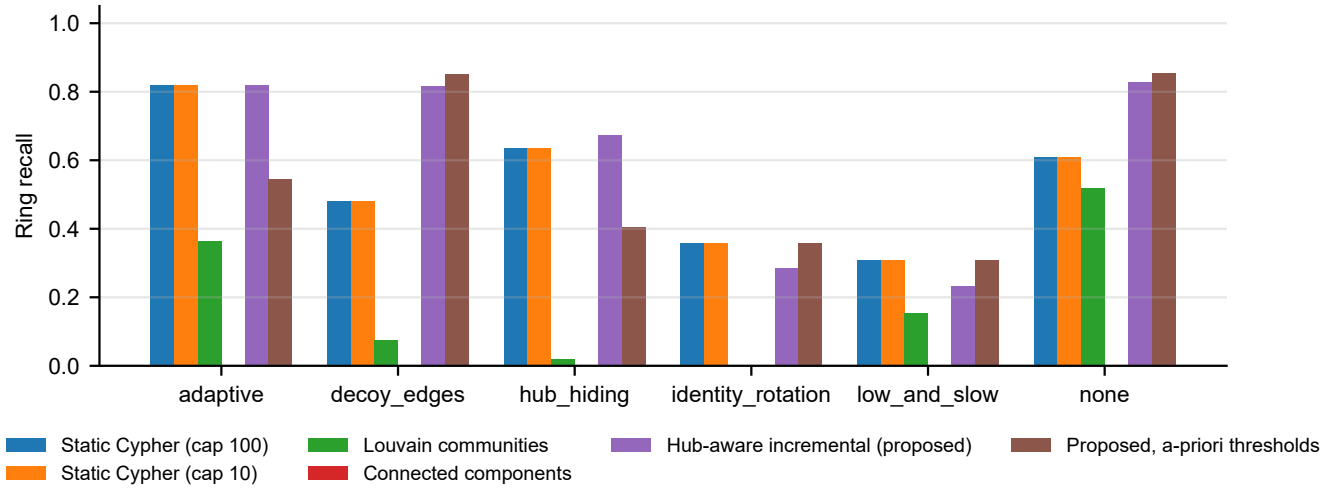



Fig. 7. Ring recall by evasion strategy, pooled over all E1 runs.

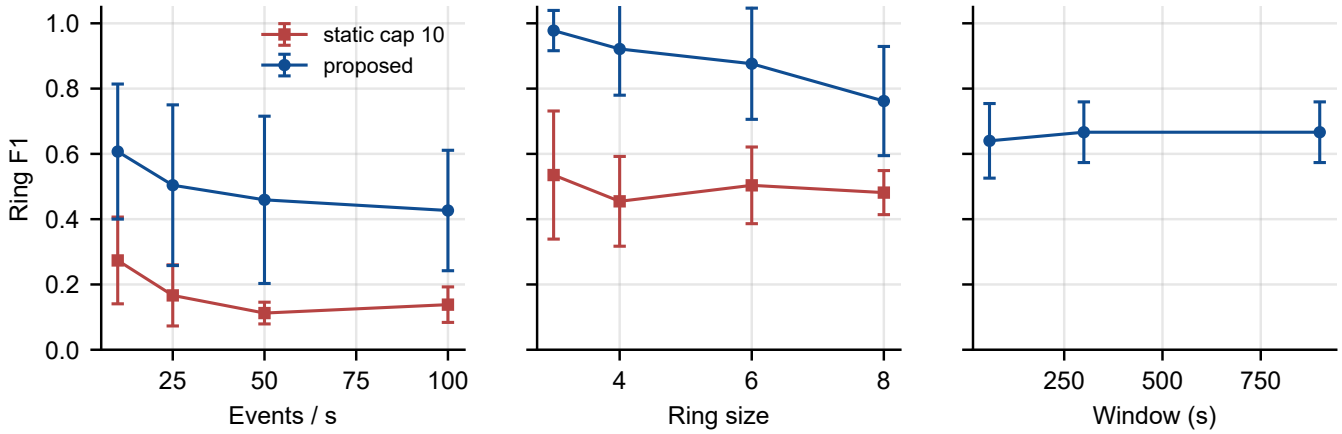



Fig. 8. Ring F1 of the proposed method and of the static rules (cap 10) when the event rate (`adversarial`), the planted ring size (`baseline`) and the view window (`drift`, proposed only) change; 5 test seeds per point.

last 60–90 s, short against the 300 s half-life, so decay has little to act on here; its value on long streams is untested.

### *F. Public data*

None of the public datasets we could obtain has device, card or IP links, so on all of them only the transaction detectors (temporal cycle, smurfing) run, and the results below say nothing about hub-aware weighting (Table VI; sources and hashes in `docs/datasets.md`).

**AMLworld** [29] (HI-Small) is the only one with group labels: 370 laundering attempts of eight typologies. We replayed the first four days (2,284,182 transactions, 399,735 accounts) with a 1 h window and used the 55 attempts that lie entirely in that slice as ring ground truth. The detectors confirmed 303 groups; attempt recall is 0.00 and account-level precision and recall are 0.03 and 0.01. In other words, the transaction rules matched none of the attempts, and the groups they did confirm were almost all benign. Attempts span a median of 75 h, far longer than the window. A one-day window was too slow on our test machine (throughput fell steadily as the window filled), so attempts whose transfers are hours apart cannot be matched; and AMLworld's typologies (fan-in, scatter-gather, bipartite, stacks) are mostly not the relayed cycles and structuring our two transaction rules look for. The result is a negative one for short-window transaction rules on this data, not a comparison with published AMLworld models, which classify individual transactions.

**Elliptic**, as used in [25], [26], links each Bitcoin transaction to the later transactions spending its outputs, so its graph has no cycles, and amounts are not published. Neither transaction rule can fire: no account was flagged (0 of 203,769). We report it as not applicable to group detection rather than as a score.

**AMLSim** samples (IBM, 20,000 accounts, 200 planted cycles and/or fan-ins) were not flagged either (0, 0 and 0 accounts). The planted cycles are not time-ordered: none of

TABLE V
ABLATIONS ON ADVERSARIAL, HUB_HEAVY, HIGH_NOISE, BASELINE ($\times$ 5 TEST SEEDS). $\Delta$F1: PAIRED MEAN DIFFERENCE TO THE FULL SYSTEM; $d_z$: PAIRED EFFECT SIZE OF THE FULL SYSTEM OVER THE VARIANT; $p$: WILCOXON SIGNED-RANK.

| Variant | Precision | Recall | F1 | $\Delta$F1 | $d_z$ | $p$ | Latency (ms) |
|---|---|---|---|---|---|---|---|
| Full system | 0.75 $\pm$ 0.12 | 0.73 $\pm$ 0.12 | 0.72 $\pm$ 0.11 | – | – | – | 19.2 |
| w/o hub weighting | 0.65 $\pm$ 0.15 | 0.58 $\pm$ 0.14 | 0.60 $\pm$ 0.14 | -0.12 | 0.42 | 0.161 | 24.0 |
| w/o time decay | 0.80 $\pm$ 0.11 | 0.72 $\pm$ 0.10 | 0.75 $\pm$ 0.09 | +0.02 | -0.18 | 0.277 | 17.6 |
| w/o incremental scheduling | 0.75 $\pm$ 0.12 | 0.73 $\pm$ 0.12 | 0.72 $\pm$ 0.11 | 0.00 | – | – | 18.4 |
| w/o risk model | 0.28 $\pm$ 0.09 | 0.82 $\pm$ 0.09 | 0.39 $\pm$ 0.10 | -0.33 | 2.05 | $<$0.001 | 15.1 |
| w/o hysteresis | 0.72 $\pm$ 0.12 | 0.74 $\pm$ 0.10 | 0.70 $\pm$ 0.10 | -0.02 | 0.37 | 0.154 | 22.7 |
| w/o hub-aware detector | 0.98 $\pm$ 0.05 | 0.25 $\pm$ 0.09 | 0.36 $\pm$ 0.12 | -0.36 | 1.02 | 0.001 | 5.3 |
| w/o cycle detector | 0.75 $\pm$ 0.14 | 0.60 $\pm$ 0.11 | 0.64 $\pm$ 0.11 | -0.09 | 1.02 | 0.002 | 15.4 |
| w/o smurfing detector | 0.72 $\pm$ 0.14 | 0.64 $\pm$ 0.15 | 0.65 $\pm$ 0.14 | -0.07 | 0.93 | 0.003 | 19.9 |

the 21 we could reconstruct from equal hop amounts has its hops in time order (one cycle's five hops fall at steps 28, 113, 84, 59 and 78), whereas our cycle detector, by design, looks for money relayed around a loop.

## VI. DEPLOYMENT, CALIBRATION AND EXPLANATION STUDIES

The experiments in this section were added after the main evaluation. Each has a protocol committed to the repository before its first result (`docs/research/`), and every amendment is recorded there with its reason.

### A. RQ6: latency on the deployed stack

The latencies of Section V come from the in-memory store. We therefore replayed the paced E1 scenarios and the users sweep through the deployed path: Redis Streams into the detection engine, with the graph in Neo4j 5.26. All 52 runs produced the same sequence of ring updates and the same flagged rings as the in-memory engine (52 of 52), so the store and the bus change timing, never results. Paced at simulated real time, a flag followed the event that completed its batch by a median of 176–438 ms across the five scenarios. Each detection reads its region through Cypher round trips and took 3 to 19 times as long as on the in-memory store for the same batches. When events arrive faster than the store can take them, Neo4j sets the ceiling: 35–52 events per second on the test laptop, with events queueing in Redis and flags arriving 9–19 s after their event. Varying the consumer's read size and blocking timeout, after HYADES [37], changed nothing measurable at these rates. Neo4j saturates before the consumer does.

### B. RQ7: ring-level calibration and early warning

The scorer's isotonic calibration was fitted on snapshot rows, but analysts read scores of ring-level alerts. On 1383 ring-level units from the test seeds its expected calibration error is 0.10, and rings scored in the 0.8–0.9 and 0.9–1.0 bins were true rings 29% and 74% of the time. A calibrator refitted on ring-level units of separate seeds, in the manner compared by Azbergen et al. [32] and Ding et al. [33], lowers the Brier score from 0.085 to 0.044. Ranking holds up better than calibration: precision at the top 5% of units is 0.79. Early warning, measured without look-ahead, is rare. Of 227 true rings with a forming phase, 30 (13%) were flagged before activation, a median of 2.2 s early, and 84% of first early warnings pointed at a real ring.

### C. RQ8: explanation quality

We measured the explanations the product shows on the 262 alerts of the test streams. Writing the unit test for counterfactual validity exposed a defect: counterfactuals aimed at the raw score of the calibrator's breakpoint, but the calibrator interpolates between breakpoints, so the proposed change left the calibrated score above the threshold. As shipped, 0 of 103 counterfactuals flipped the decision. Aimed at the interpolated crossing, 103 of 103 do. The correction was registered as a protocol amendment before the experiment ran. Any system that derives counterfactual targets from an isotonic calibrator can make the same mistake.

Coverage is the larger limitation: 69% of alerts have no single-feature counterfactual over the actionable features, because typical transfer size dominates their score. The explanations are stable. Under 5% noise in non-actionable features, after Artelt et al. [34], the contribution vector moves by a median of 0.026 (relative $L_1$), and the top three features never change for the median alert. A scorer retrained on disjoint seeds, after Movin et al. [35], ranks the contributions similarly (median Spearman 0.92) and still raises 87% of the alerts. It proposes the same first counterfactual for only 29%, though, so a counterfactual describes this model, not the ring. Against simulator ground truth, the evidence subgraph of a correct alert is the true ring: median precision 1.00 and coverage 1.00.

### D. RQ9: a learned baseline and evasion strength

To test whether the rule stage is needed, we trained gradient boosting on the same 19 ring features over rule-free candidate groups (Louvain communities without a density filter). Training, calibration and threshold selection used the shipped scorer's protocol. On E1's test streams it reaches F1 0.555 against 0.731 (paired Wilcoxon $p$ ¡ 0.001, rank-biserial -0.85). Precision is not the problem: it is 0.94 against 0.76, with a third of the alerts. Recall is: 0.38 against 0.74. The gap is largest where hubs and evasion matter (`hub_heavy`: 0.40 against 0.79; `adversarial`: 0.24 against 0.59).

Turning up camouflage strength, after Liu et al. [4] and Hu et al. [8], separates the methods further. With up to eight decoy

TABLE VI

PUBLIC DATA, TRANSACTION DETECTORS ONLY (NO DEVICE/CARD/IP LINKS EXIST). ACCOUNT PRECISION/RECALL: ACCOUNTS IN CONFIRMED GROUPS VS. ACCOUNTS IN LAUNDERING TRANSACTIONS. ATTEMPT RECALL: SHARE OF LAUNDERING ATTEMPTS THAT LIE ENTIRELY IN THE SLICE AND ARE AT LEAST HALF COVERED BY ONE FLAGGED GROUP.

| Dataset | Slice | Rows | Attempts | Acc. precision | Acc. recall | Attempt recall |
|---|---|---|---|---|---|---|
| amlsim_20K_cycle200 | all | 117,805 | 0 | – | 0.00 | – |
| amlsim_20K_fanin200 | all | 118,250 | 0 | – | 0.00 | – |
| amlsim_20K_fanin200cycle200 | all | 120,558 | 0 | – | 0.00 | – |
| elliptic | all | 234,355 | 0 | – | 0.00 | – |
| amlworld | 2022-09-01 to 09-05 (excl.) | 2,284,182 | 55 | 0.03 | 0.01 | 0.00 |
| attempt recall by typology: bipartite 0.00, cycle 0.00, fan-in 0.00, fan-out 0.00, gather-scatter 0.00, random 0.00, scatter-gather 0.00, stack 0.00 | | | | | | |

edges per ring action, the proposed method's recall of the decoy rings stays between 0.74 and 0.82, while Louvain falls to 0.00 and the learned baseline to 0.05. Slow rings remain the weakness found in RQ3: the proposed method recalls at most 0.25 of them, and static Cypher with a degree cap of ten recovers more (0.42–0.75), though at a precision near 0.04. In the 60 s scenario, strong slow and identity-rotation settings produce few rings, so those two sweeps have little statistical power. We report them as indicative.

## VII. THREATS TO VALIDITY

**Simulator realism.** All main results come from our simulator. Hub sizes, ring behaviour and evasion strategies follow our configuration, not observed fraud. A detector tuned to this generator may do worse on real traffic, and the absolute numbers should not be read as field performance.

**Synthetic labels.** Ground truth is the set of rings the simulator planted. Benign groups that happen to look like rings count as false positives, and planted rings that never become visible in the stream still count against recall; both directions affect every method equally.

**Public data.** The public datasets we could obtain carry no device, card or IP links, so they cannot test hub-aware weighting at all; on them only the transaction detectors run (Section V-F). Only AMLworld has group labels, and we could replay only a four-day slice with a one-hour window. Elliptic and the AMLSim samples turned out to be outside what the transaction rules can detect, for structural reasons we describe. A real-world test of hub-aware weighting would need data with shared-attribute links, which we did not find in public form.

**Tuning.** Thresholds were selected on validation seeds of the same generator; the test seeds are independent draws but from the same distribution. In the weight-function comparison, three of the five selected edge thresholds lie on the edge of the grid, so those functions might do somewhat better with a wider grid.

**Scale.** The largest graph we measured has 100,000 users. A million users was not feasible on the 7.7 GB test machine: at 100,000 users less than 0.3 GB of memory was free and one full recomputation already took over 20 s per batch. In the scale scenario, attribute counts grow with the number of users while hub sizes stay fixed; graphs whose hubs grow with the population would give larger dirty regions. The 250-user first-hop limit was tested with hubs of up to 10,000 accounts, five seeds per size, on one scenario; its harmlessness rests on the hub-aware weight keeping a single busy attribute below the edge threshold, and would not hold for a weight or threshold that does not.

**Latency.** The latencies of Section V come from the in-memory store on a laptop. Section VI repeats the measurement on Neo4j and Redis Streams. The detected rings are identical, but detection is several times slower and throughput is bounded by the store. Both figures describe one laptop running Docker, not a server.

**Baselines.** The scored baselines of Table II use the shipped scorer, which was trained on candidates of the proposed pipeline; a scorer trained on each baseline's own candidates might do better for it. The learned baseline of Section VI is a tabular model over Louvain groups, so its weakness under decoy edges is partly the grouping's. We did not run a graph neural network. A GraphSAGE node classifier with a fixed rule for turning node scores into rings, as used by Badjie et al. [25] and Kumar et al. [26], is the obvious next comparison, together with node-level AUC on public benchmarks. Our node-level numbers are not comparable to those benchmarks.

**Calibration.** The shipped scorer is miscalibrated at the ring level (Section VI). Its scores rank alerts well but should not be read as probabilities until a ring-level calibrator replaces it.

## VIII. CONCLUSION

Keeping links through busy shared attributes, instead of capping them, lets a streaming detector reach a recall on rings that share busy infrastructure that no cap reaches. The shape of the discount matters little: a constant weight with its own threshold did as well as $1/\ln(1+d)$, while strong discounts and soft caps lost the rings hidden behind hubs. On its own the weighting is not precise, and a calibrated scorer over ring features supplies most of the precision. Given the same scorer, the degree-capped rules reach F1 0.56 against 0.73 for the full pipeline, so the scorer accounts for most of the gain over the original rules, and at the best operating point on shared-attribute rings the weight and the cap are not distinguishable on F1. Restricting detection to the region a micro-batch touches gave the same rings as full recomputation in every accuracy run, and kept per-batch latency flat as the graph grew to 100,000 users while full recomputation grew past 20 s. Rings that rotate identities or act slowly evade all methods we tested, and all of our positive results are on synthetic data. On Neo4j

and Redis Streams the detector flags exactly the same rings, at the cost of slower detection and a throughput bounded by the store. A learned model over rule-free groups is more precise but finds half as many rings, and it collapses under decoy edges, which the hub-aware rules withstand. The explanations are stable and their evidence is faithful. Measuring them also found and fixed a calibration defect that had made every counterfactual invalid. The next steps are data with real shared-attribute links, a ring-level calibrator, a GraphSAGE baseline, multi-feature counterfactuals, and longer runs for slow and identity-rotating rings.

Code, scenarios, models and the scripts that regenerate every table and figure are in the project repository (`REPRODUCE.md`).

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